
BIOLOGICAL TREATMENT METHODS



ACTIVATED SLUDGE PROCESS

- Basic activated sludge process / conventional aeration process the recirculated activated sludge is added to the inlet end of the aeration tank as a single dose
- Regime flow employed in the aeration tank is plug flow and not mixed flow
- Plug flow implies that the sewage moves down progressively along the aeration tank, essentially unmixed with the rest of the tank contents
- Other type of flow regime called complete mixed flow involves the rapid dispersal of the incoming sewage throughout the tank and is adopted in the extended aeration process
- In conventional aeration tank, the F/M ratio and the oxygen demand will be the highest at the inlet end and will then progressively decrease.
- In complete mix system on the other hand, the F/M ratio and the oxygen demand will be uniform throughout the tank

ACTIVATED SLUDGE PROCESS CONTD....

- Plug flow regime is achieved in such an activated process by employing a long and narrow configuration of the aeration tank with a length equal to 5 to 50 times the width
- The sewage and the mixed liquor are let in at the head of the tank and withdrawn at its end. Because of plug flow regime, the oxygen demand at the head of the aeration tank is high and then tapers down. However, air is supplied in the process at a uniform rate along the length of the tank
- Leads to either oxygen deficiency in the initial zone or wasteful application of air in the subsequent reaches
- Conventional system consists of an aeration tank. Plant consists of aeration tank, a secondary settling tank, and a sludge return line and an excess sludge waste line leading to digester. BOD removal 85-92%

LIMITATIONS OF CONVENTIONAL ASP

- Aeration tank volume requirement is high
- There is a lack of operational stability at times of excessive variations in the rate of inflow or its BOD strength
- To overcome such difficulties posed by a conventional system plant, and to meet the specific treatment objectives, several modifications of conventional system have been suggested by modifying the process variables.
- In spite of its limitations- most widely used type of activated sludge process

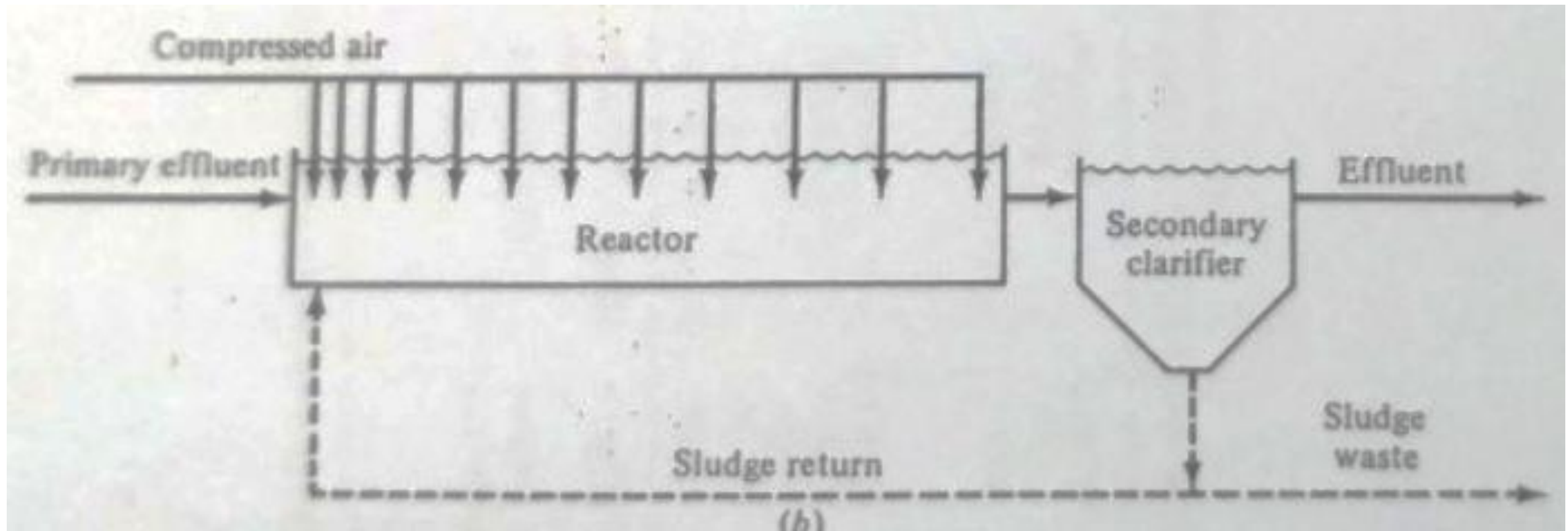
MODIFICATIONS OF ASP

- Tapered aeration process
- Step aeration process
- Contact stabilisation process
- Complete mix process
- Modified aeration process
- Extended aeration process
- Activated aeration process

TAPERED AERATION PROCESS

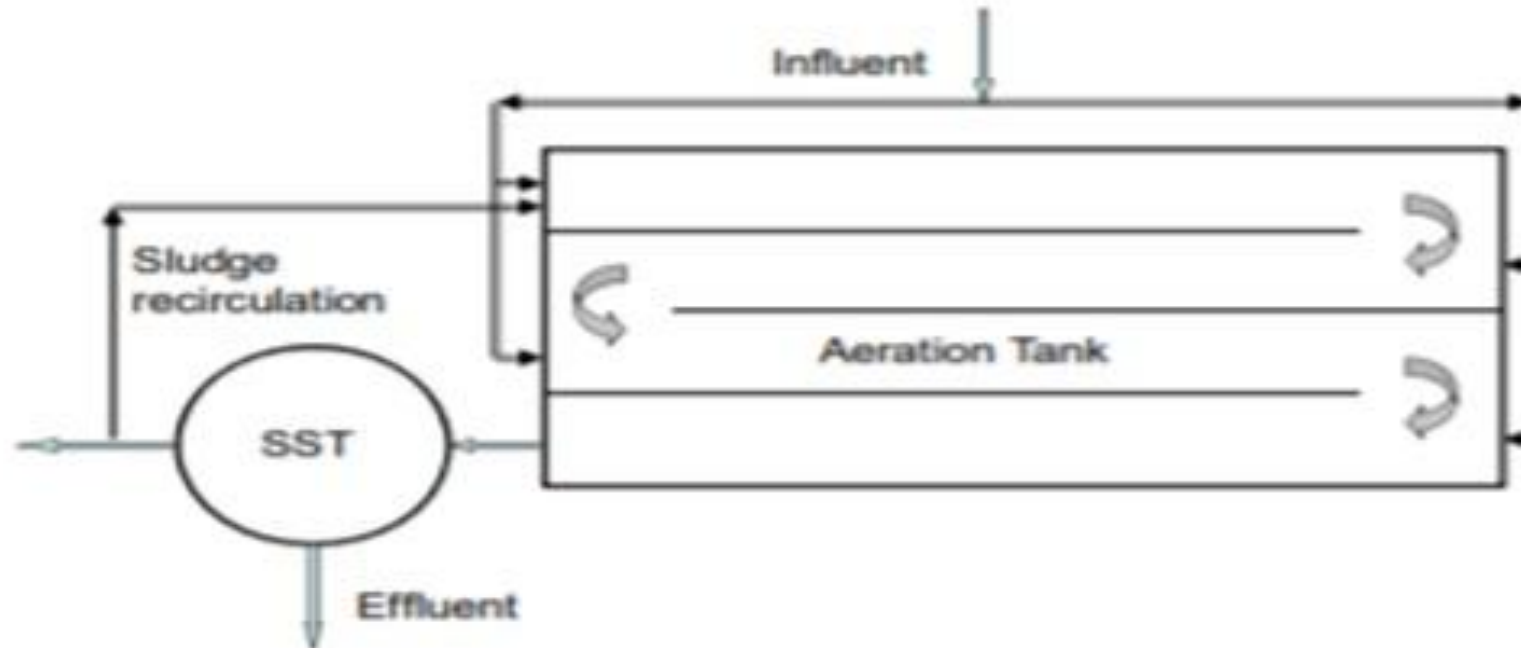
- Process involves little modification of the conventional process, and ensures higher air supply at the inlet and in the initial length of the tank, as compared to the downstream length
- As the mixed liquor progresses through the aeration tank, its air requirement goes on reducing
- In a tapered aeration plant, compressed air is supplied at higher rates near the inlet end of the tank, and is gradually decreased as the sewage moves towards the outlet end of the tank
- Such process helps us in ensuring optimal application of air in aeration tank
- No of diffuser plates are used in diffused air aeration, adopted in all modern designs

TAPERED AERATION PROCESS CONTD....



STEP AERATION PROCESS

- Sewage is introduced along the length of the aeration tank in several steps, while the return sludge is introduced at the head



STEP AERATION PROCESS CONTD....

- Arrangement results in a uniform air requirement along the entire length of the tank, and hence uniform air supply of the conventional plants, can be efficiently used
- Process enables an appreciable reduction in aeration tank volume, without lowering the BOD removal efficiency
- Step aeration process has considerable capacity to absorb shock organic loadings
- Method found applicable for large plants of capacities upto 1000MLD

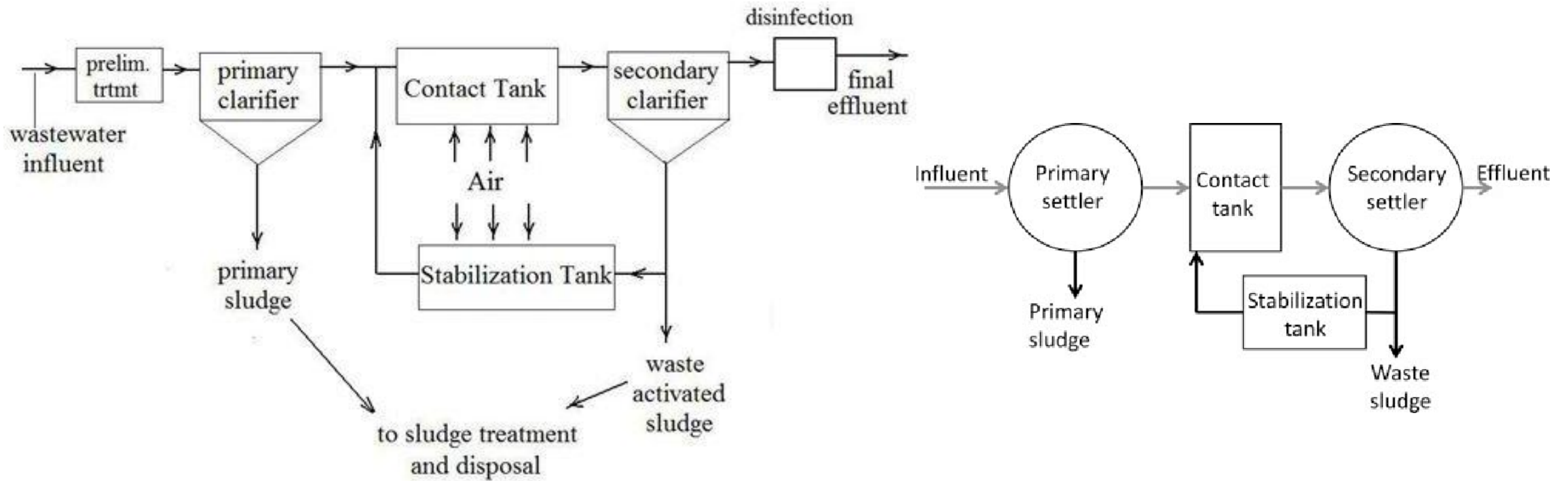
CONTACT STABILISATION PROCESS OR BIOSORPTION PROCESS

- Process suitable for treating colloidal wastewaters
- Sewage and recycled or returned sludge are mixed and aerated for a comparatively shorter period of 0.5 to 1.5 hour in a special mixing tank, called contact tank
- Mixing will allow the suspended and dissolved organic matter to be sorbed to the activated sludge floc
- Sorbed organics and flocs are removed in the secondary settling tank, where the effluent from the contact tank enters
- These settled sorbed organics and flocs are then transformed to a sludge aeration tank called stabilisation tank or aerodigester where the organics are stabilised over a period of about 3 to 6 hours before it is fed back into the contact aeration tank
- Stabilised sludge is then mixed with the influent wastewater again, and the process is repeated

CONTACT STABILISATION PROCESS CONTD...

- Effective in removal of colloidal and suspended organic matter , but not very effective in removing soluble organics
- Well suited for domestic sewage , containing low percentage of soluble BOD
- Compared to the conventional system, this process has greater capacity to handle shock organic loadings, because of biological buffering capacity of the sludge reaeration tank
- Process also presents greater resistance to toxic constituents only for a short time
- Air requirements – same as that of conventional ASP

CONTACT STABILISATION PROCESS OR BIOSORPTION PROCESS

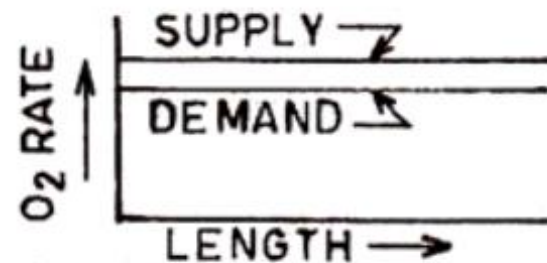
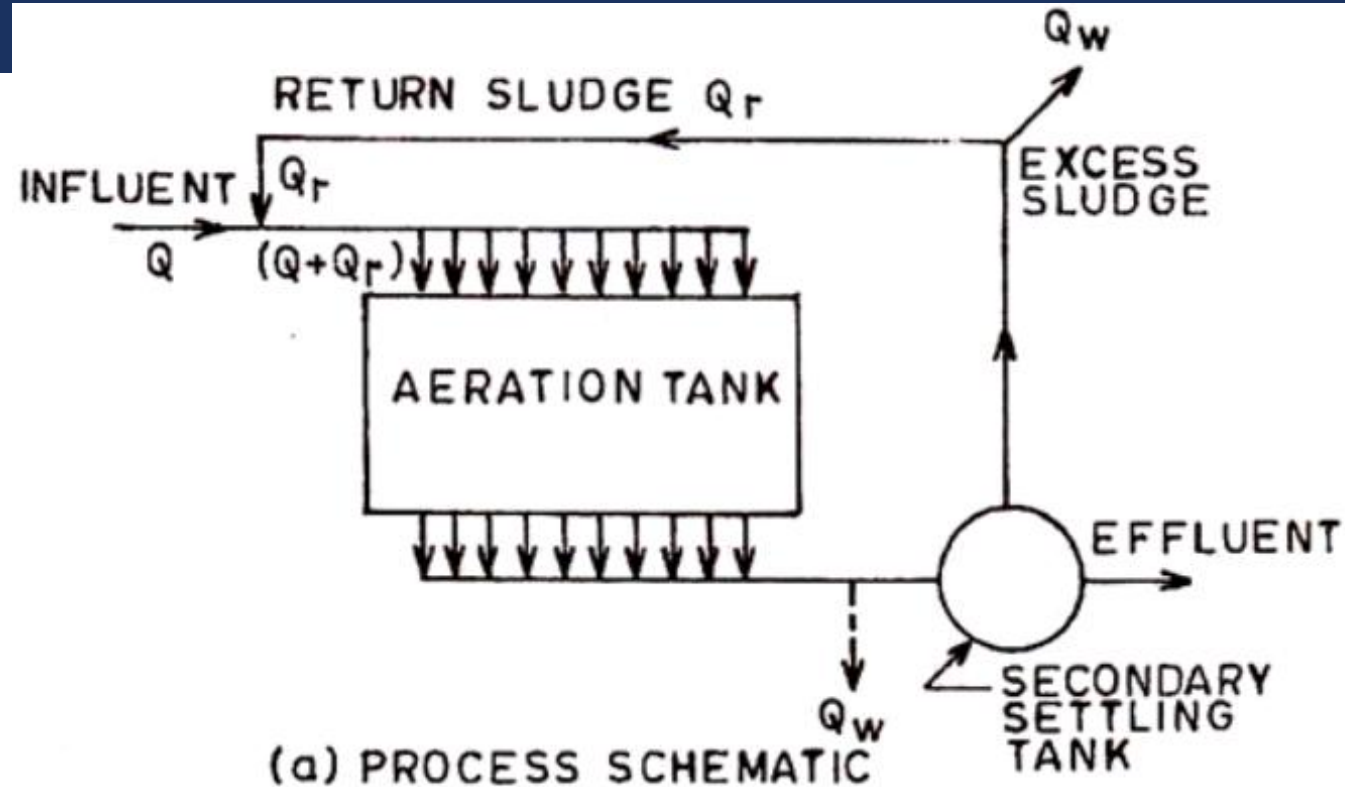


Contact Stabilization Activated Sludge Flow Diagram

COMPLETE MIX PROCESS

- Developed for small cities, where hourly variations in sewage were quite high, and as such conventional plants are experiencing serious problems of biological instability
- The plug flow regime of a conventional plant is replaced by a completely mixed flow regime
- Such flow regime can be achieved by thorough mixing of sewage and return sludge
- Sewage and return sludge are therefore distributed uniformly along one side of aeration tank , and the aerated sewage is withdrawn uniformly along the opposite side
- Mechanical aerators are installed in the centre of a circular tank , which may be used in such a plant
- Such mechanical aerators must have adequate mixing capacity to ensure thorough mixing of sewage and return sludge

COMPLETE MIX PROCESS



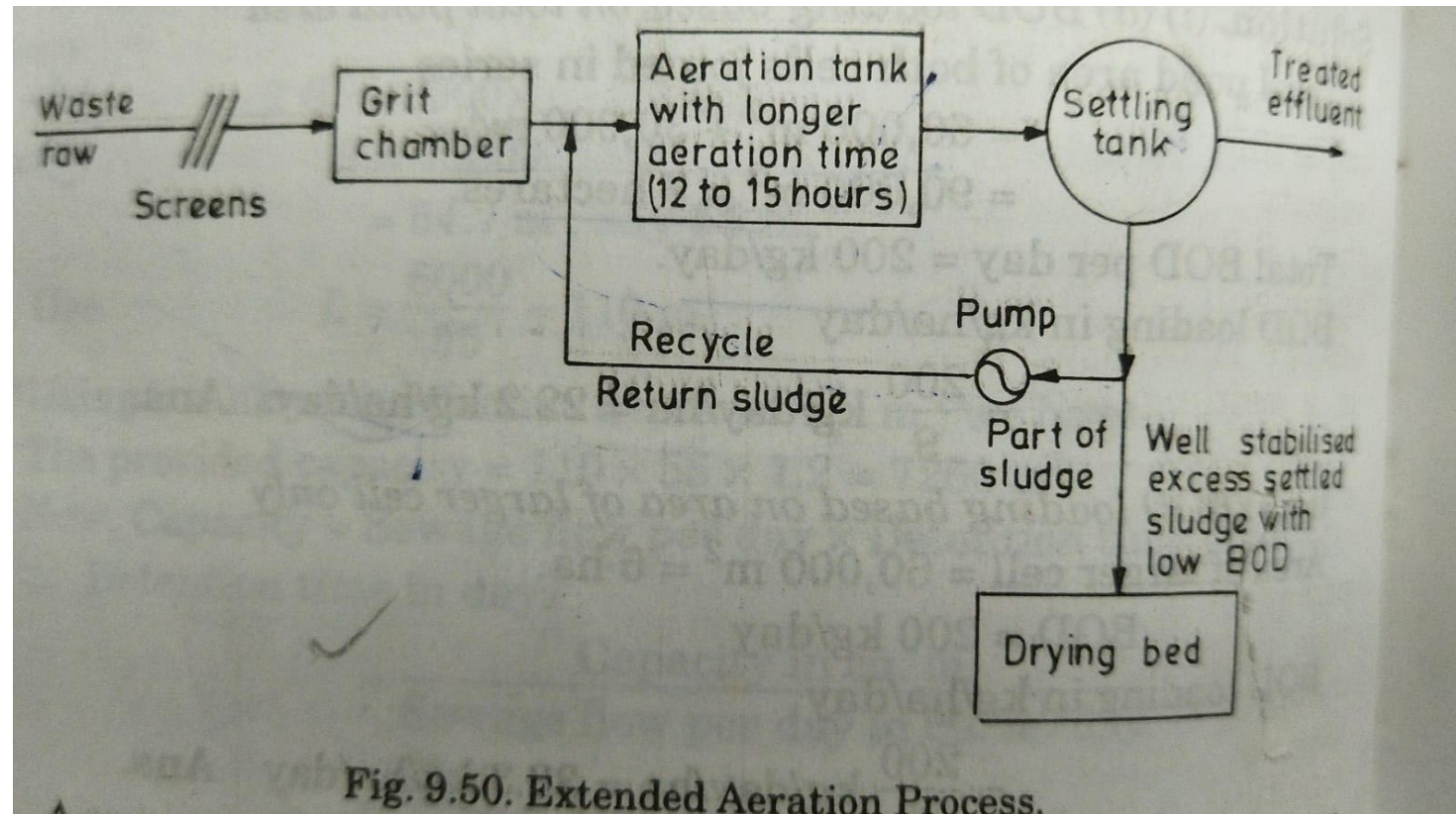
MODIFIED AERATION PROCESS

- When an intermediate quality of effluent containing higher BOD is permissible, such as at a place where effluent is to be used for farming, this modified aeration plant may be adopted
- Leads to substantial savings in construction and aeration costs
- Doesnot need any primary sedimentation tank
- Short aeration period , high volumetric loading, High F/M ratio, low percentage of sludge return, low concentration of MLSS
- BOD removal - 60-75%

EXTENDED AERATION PROCESS

- Similar to that of a complete mix process
- Primary sedimentation – avoided , Grit chamber or comminutor is provided for screening
- Aeration process is quite large – 12-24 hours (4-6 hours in conventional plant)
- No separate sludge digester is required- solids undergo considerable endogenous respiration and get well stabilised over the long detention periods
- Thus the sludge can be directly taken to sludge drying beds
- Operation is simple due to the elimination of primary settling and separate sludge digestion
- Applicable for small communities having flow less than 4MLD

EXTENDED AERATION PROCESS



ACTIVATED AERATION PROCESS

- A pair of conventional ASP operating in parallel
- Excess activated sludge from secondary settling tank of one unit will supply to the aeration tanks of both the units and sludge from the second unit is pumped to final disposal
- No modification in conventional process, however reduces the cost of construction costs, operating costs

BIO TOWERS

- Sometimes known as trickling filter – one of the two main types of biological control units
- Industrial and municipal WWTP use bio-towers as part of the secondary treatment process
- Bio-tower is a round tank structure and contains usually engineered growth media with high surface area
- Rock, slag and wood was a common growth media in the past but they provide lower surface area, limiting the hydraulic flow rate of the bio-tower so materials such as Polyvinyl chloride, polypropylene with a defined surface area are utilised



Bio tower at Madras Refineries Ltd, Chennai

BIO TOWERS CONTD....

- The bio-tower takes up less room because it is fitted with PVC structured media, which offers a bigger surface area of about $100\text{m}^2/\text{m}^3$
- When two towers are connected in series , a bio-tower system can readily reduce BOD by 60 to 80%
- Skeletal civil construction is sufficient because there is no lateral stress of media and it is self supporting, costs for building are reduced as a result
- Low power consumption

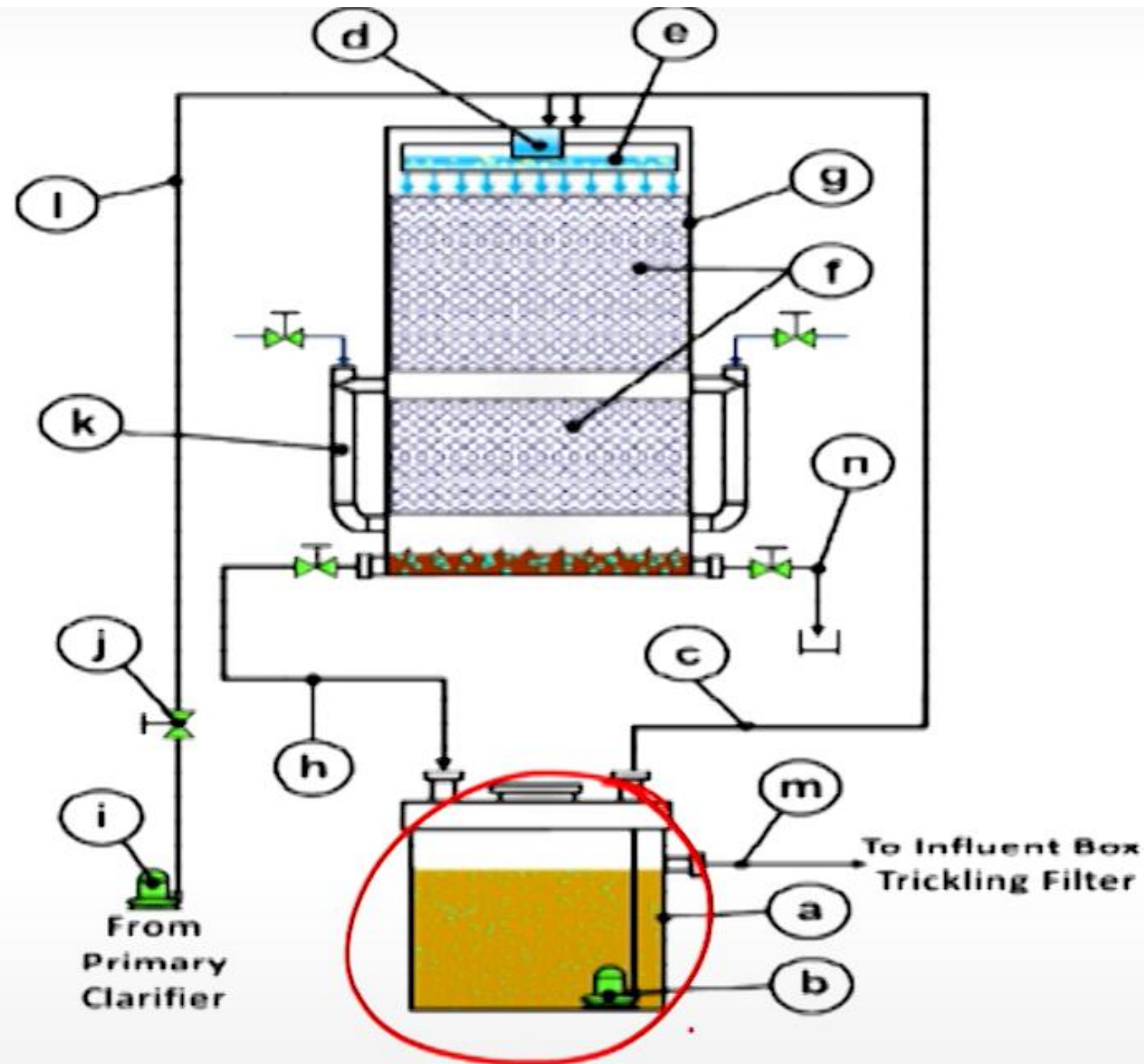
BIO TOWERS , WHERE IT CAN BE USED ?

- When a multi-stage biological treatment process is required to break down complex organics, a bio-tower can be used
- Bio-tower can be used in situations where the system has to deal with shock loads from both hydraulic and organic disturbances

FUNCTIONING

- In bio-towers (aerobic or anaerobic) , a distribution system equally spreads the waste water over a biofilm covered growth media
- The wastewater then trickles over the medium and is collected at the bottom of the bio-tower. The growth media is located above the bottom of the bio-tower to ensure air transfer throughout the filter media
- A pump lifts the inflowing wastewater to the distributor as well as partially recirculated the effluent
- The process of removing organic matter by this method can be successfully combined with a nitrification process
- In the upper portions the heterotrophic bacteria outgrow the nitrifying species
- As soon as the organic matter in the waste water is subsequently decreased below a threshold concentrated , the nitrifying bacteria can complete and initiate nitrification

BIOTOWER



Bio-Tower System: a) Reservoir, b) Pump, c) Recirculating Pipe, d) Influent Pan, e) Distributor f) Growth Media, g) Bio-tower vessel, h)Return pipe, i) Feed pump, J) Flow valve, k) Air distributor, l) Feed pipe, m) Discharge pipe, n) Drain port

BIOTOWER

- Aerobic bacteria clings to the surface of media and stays in the system in the form of biofilm organic matter in waste water comes in contact with the bacteria consumes the organic matter and releases carbon dioxide, water and humus
- As the layer grows in thickness, bottom layers turn into anaerobic because of lack of oxygen also organic matter does not reach bottom layers and bacteria goes into endogenous phase and gets detached from media. This is called Sloughing
- Lower residual suspended solids concentrations are observed in treated wastewater
- A bio-tower is a cost effective addition to other treatment methods
- Improved plastic media design, such as cross flow media, has significantly reduced the size and cost of bio-towers while increasing their efficiencies
- Mechanical or diffuse aeration is not required, this works on natural draft

DISADVANTAGES BIOTOWER

- Low loading rates and large reactor volumes
- Stripping of volatile organic chemicals
- Release of odours

APPLICATIONS

- Municipal Sewage
- Pharmaceuticals
- Distillery
- Paper mill
- Paint industry
- Food Processing industry
- Dairy
- Sugar Brewery

STABILISATION POND

- The ***stabilisation ponds*** are open-flow through basins designed and constructed to treat sewage and biodegradable industrial wastes.
- They provide long detention periods extending from a few to several days.
- During this time the waste get stabilized by the action of natural forces

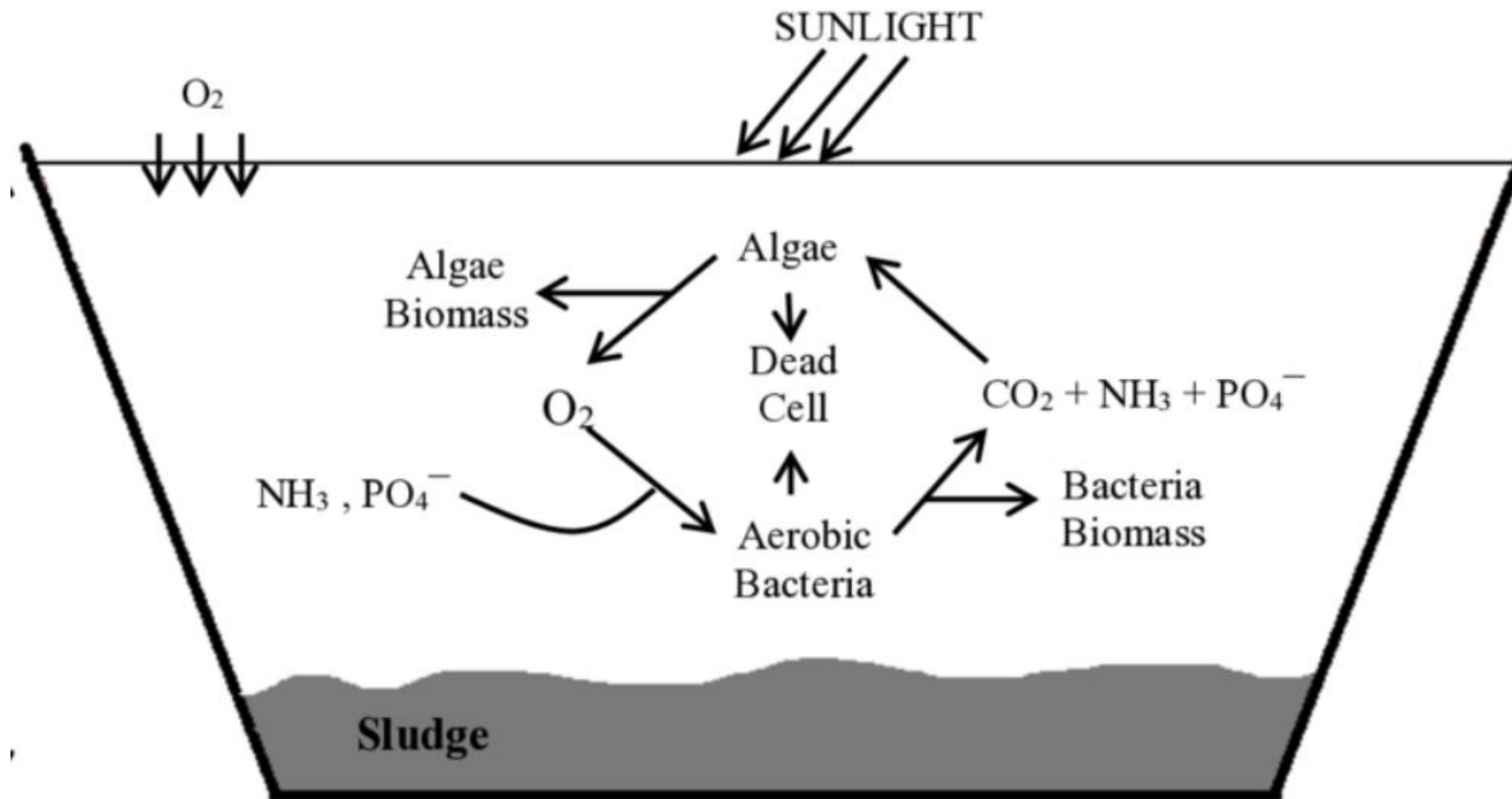
CLASSIFICATION OF STABILIZATION PONDS

- Stabilization ponds may be aerobic, anaerobic or facultative.
 - *Aerobic ponds*
 - *Anaerobic ponds*
 - *Facultative pond*

CLASSIFICATION OF STABILIZATION PONDS

- *Aerobic ponds* -are shallow ponds with depth less than 0.5 m and BOD loading of 40-120 kg/ha.d so as to maximize penetration of light throughout the liquid depth. Occasional stirring
- Stabilisation of wastes by aerobic bacteria , which flourish in the presence of oxygen
- Oxygen demand of such bacteria is met by combined action of algae and other microorganisms, called algal photosynthesis or algal symbiosis
- In this process, algae while growing in the presence of sunlight , produce oxygen by the action of photosynthesis
- This oxygen is utilized by the bacteria for oxidizing the waste organic matter
- End product – carbon dioxide , ammonia and phosphates , which are required by algae to grow and continue to produce oxygen

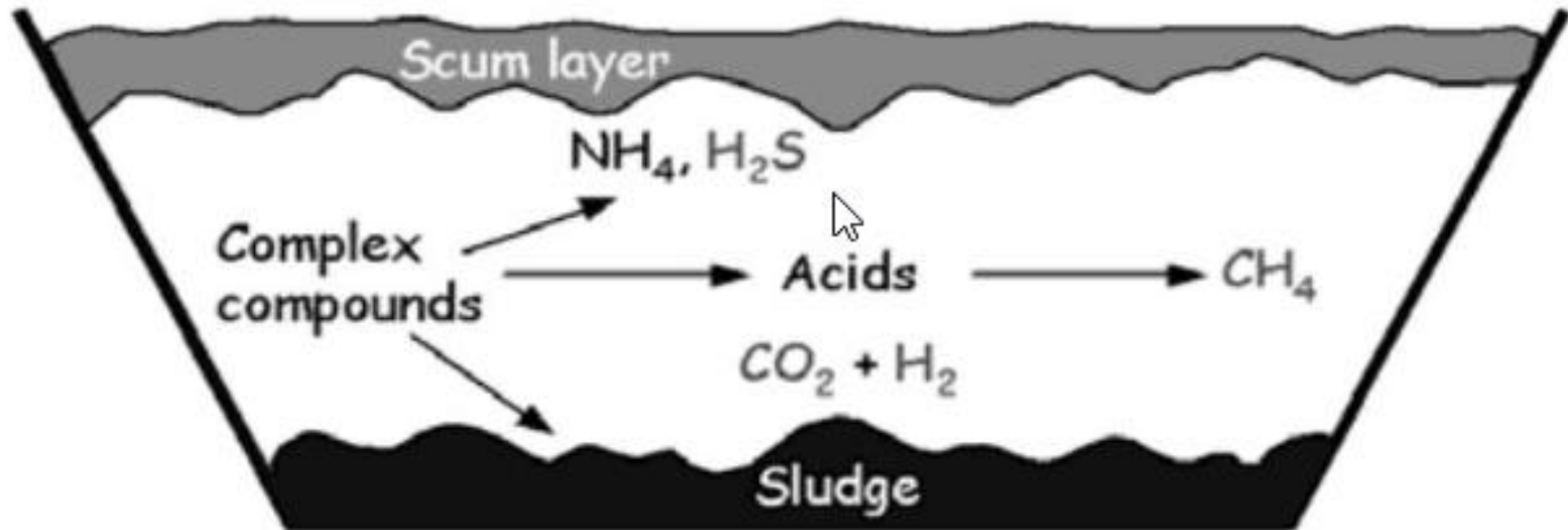
CLASSIFICATION OF STABILIZATION PONDS - *AEROBIC PONDS*



CLASSIFICATION OF STABILIZATION PONDS

- *Anaerobic ponds* are used as pretreatment of high strength wastes with BOD load of 400-3000 kg/ha.d Such ponds are constructed with a depth of 2.5-5m as light penetration is unimportant.
- Stabilisation of waste by usual anaerobic conversion of organic wastes to carbon dioxide, methane, and gaseous end products with eruption of foul odours and pungent smells

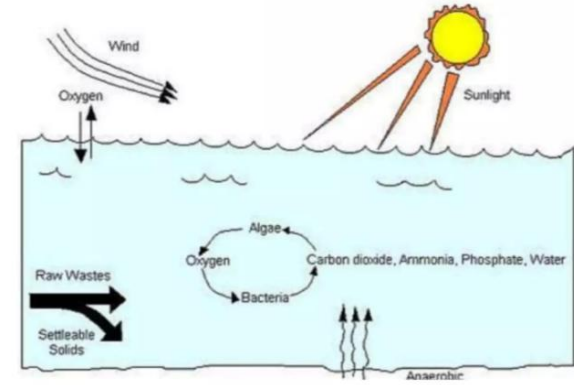
CLASSIFICATION OF STABILIZATION PONDS - *ANAEROBIC PONDS*



CLASSIFICATION OF STABILIZATION PONDS

- *Facultative ponds* They are often about 1 to 2 m in depth.
- function aerobically at the surface while anaerobic conditions prevail at the bottom.
- The aerobic layer acts as a good check against odour evolution from the pond.
- The treatment effected by such a pond is comparable to that of conventional secondary treatment process

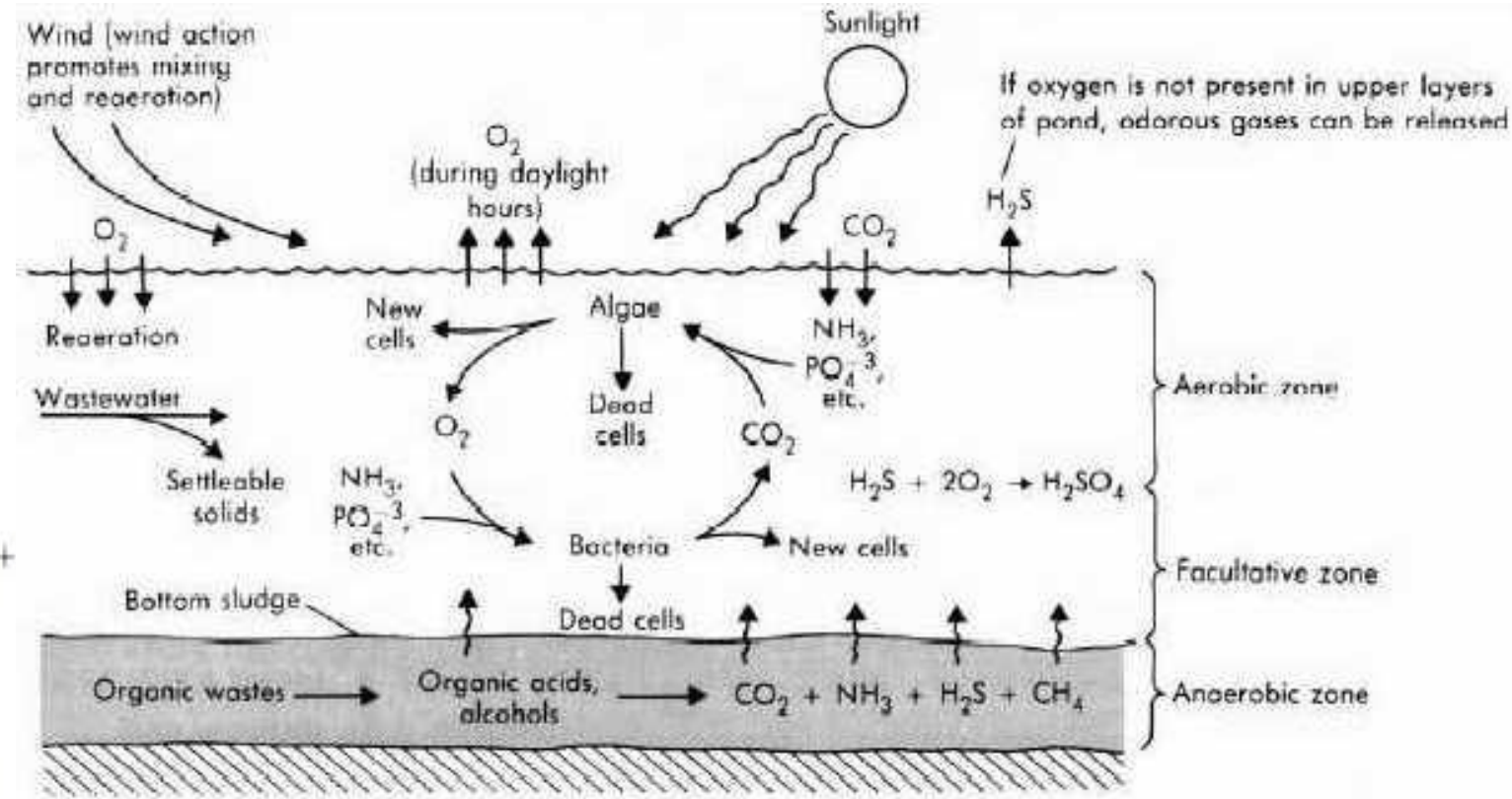
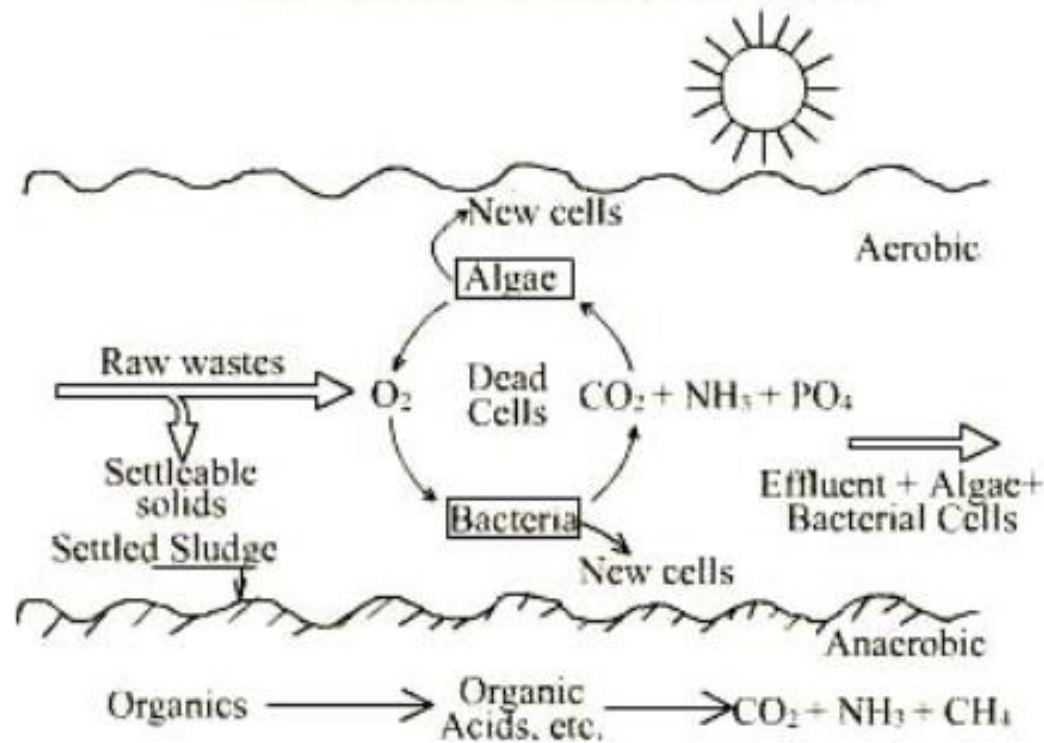
CONTD....



- The functioning of a facultative stabilization pond and the symbiotic relationship in the pond are shown below.
- Sewage organics are stabilized by both aerobic and anaerobic reactions
- In the top aerobic layer, where oxygen is supplied through algal photosynthesis, the non-settleable and dissolved organic matter is oxidized to CO_2 and water.
- In addition, some of the end products of partial anaerobic decomposition such as volatile acids and alcohols, which may permeate to upper layers are also oxidized periodically.
- The settled sludge mass originating from raw waste and microbial synthesis in the aerobic layer and dissolved and suspended organics in the lower layers undergo stabilization through conversion to methane which escapes the pond in form of bubbles.

MECHANISM OF PURIFICATION CONTD....

SYMBIOTIC RELATIONSHIP AND FUNCTIONING OF FACULTATIVE STABILIZATION POND



FACTORS AFFECTING POND REACTIONS

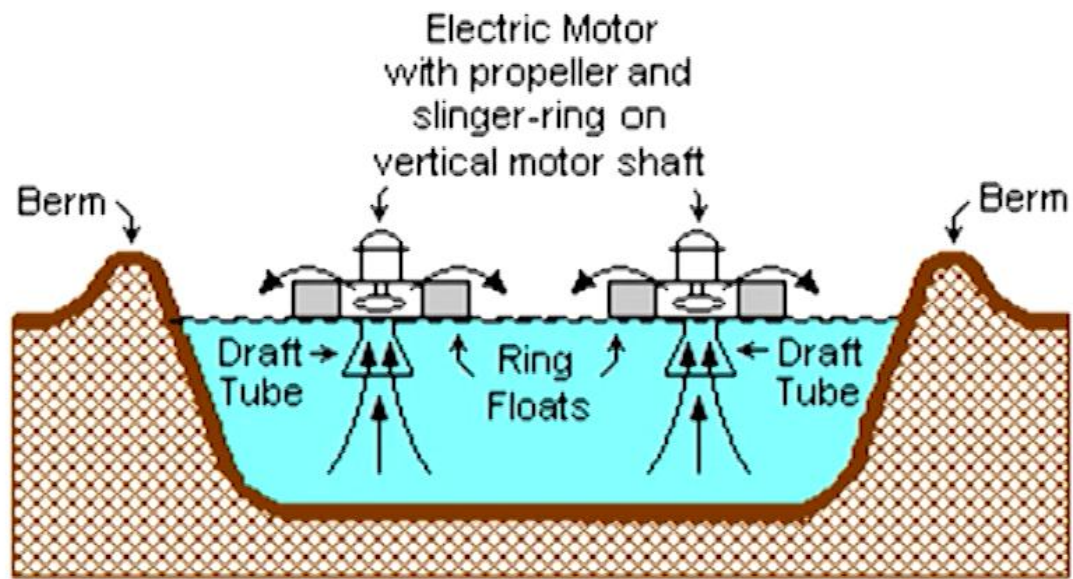
- Various factors affect pond design:
 - wastewater characteristics and fluctuations.
 - environmental factors (solar radiation, light, temperature)
 - algal growth patterns and their diurnal and seasonal variation)
 - bacterial growth patterns and decay rates.
 - solids settlement, gasification, upward diffusion, sludge accumulation.

OXIDATION POND AND STABILIZATION PONDS

- Term oxidation pond was originally referred to that stabilization pond which received partially treated sewage

AERATED LAGOON

- **Aerated Lagoon** – is a deeper oxidation pond , with oxygen introduced by mechanical aerators rather than relying on the photosynthesis oxygen production alone.
- As these ponds are deeper than the stabilization ponds , they are artificially aerated , less detention time and areas are required
- Depth of the basin ranges between 2.4 to 3.6 m , detention time 4 to 10 hours
- Frequently used for treating industrial wastewater as well as city sewage
- Based on the solid handling process and degree of mixing aerated lagoons are classified as aerobic, facultative and extended aeration system



A TYPICAL SURFACE – AERATED BASIN

Note: The ring floats are tethered to posts on the berms.

Figure 2. Aerated Lagoons (Schematic)



Figure 3. Aerated Lagoons (Actual)

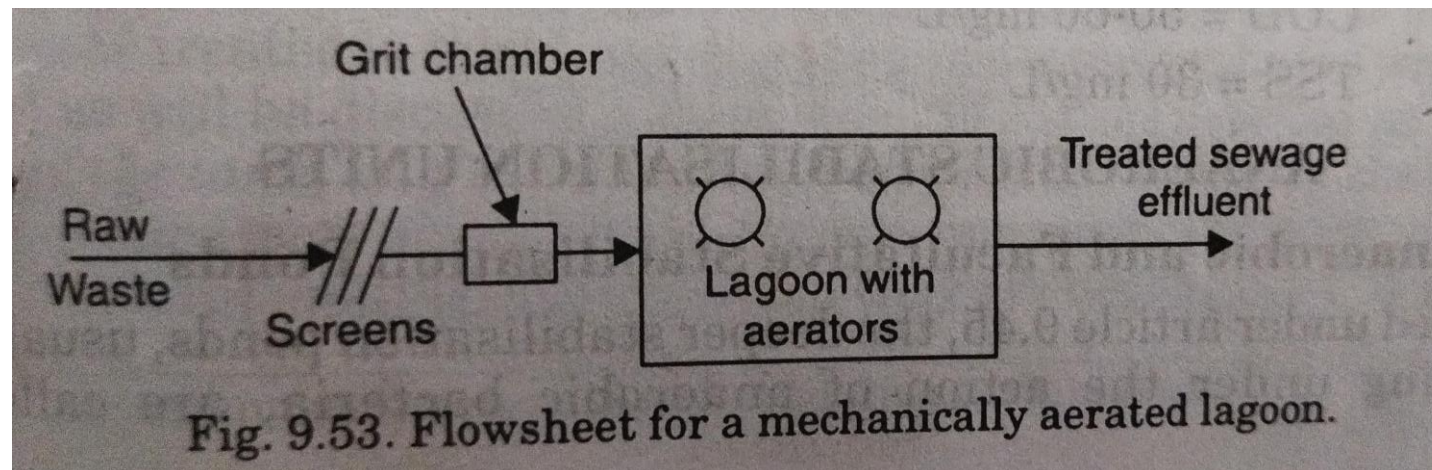


Fig. 9.53. Flowsheet for a mechanically aerated lagoon.

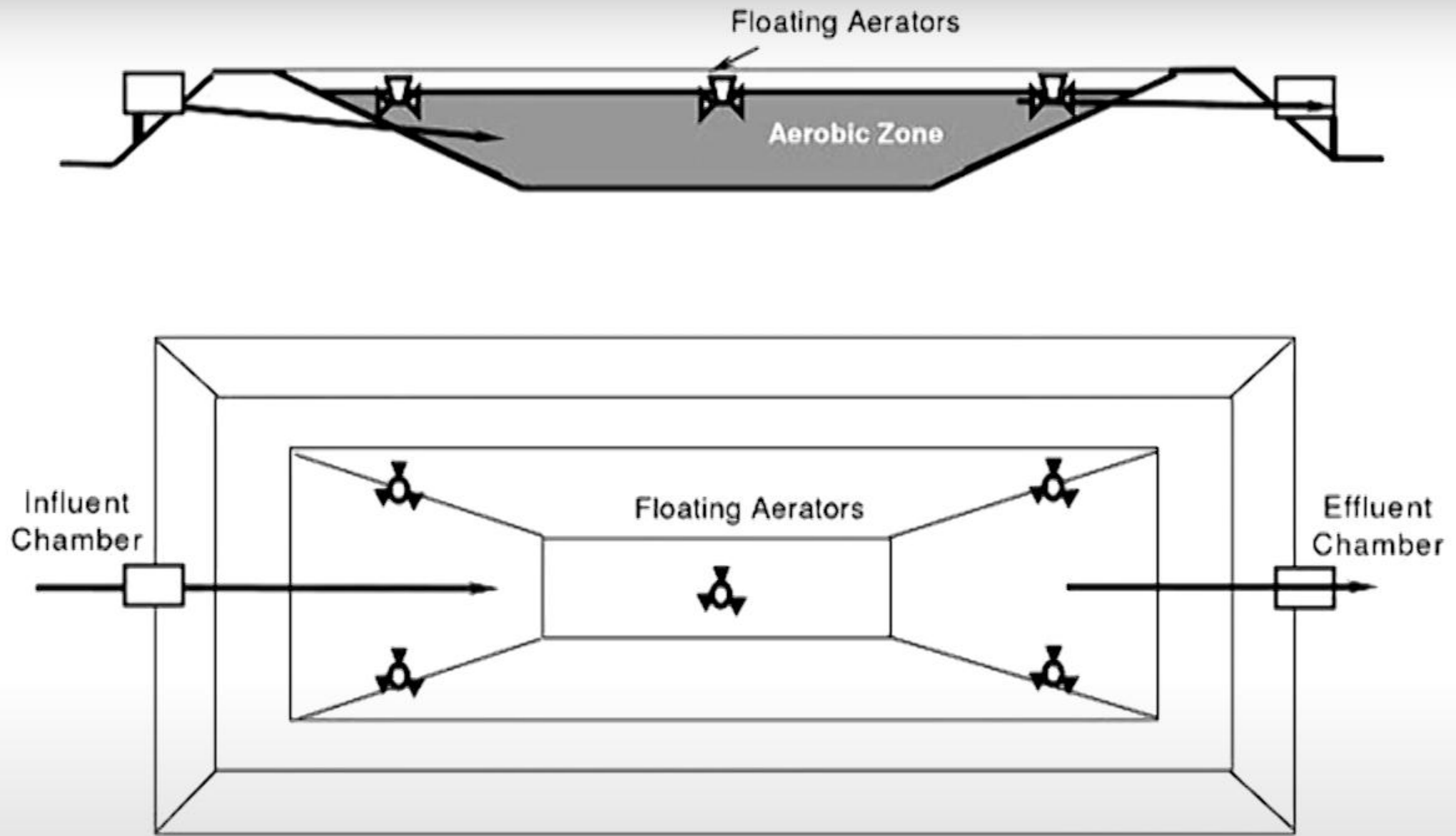


Figure 4. Plan and section of a typical mechanically aerated lagoon

AERATED LAGOON – FEATURES

Large detention time- do not have sludge recycling

A shallow depth and a large plan-view surface area – 1-3 m

A shallow depth and a large plan-view surface area

HIGH RATE ANAEROBIC TREATMENT

- Anerobic process is slow, requiring digestors of large hydraulic retention period
- Nowadays HRAT devised and constructed to treat concentrated industrial wastewater and for direct treatment of municipal wastewater
- Application of this technique is quite significant for developing countries like India, because of high energy savings and low capital and OMR costs involved in the conventional aerobic system
- Conversion of organics into methane and carbon dioxide can be obtained by maintaining a high concentration of microbes in a reactor and preventing them from escaping with the effluent- Sludge retention time(SRT)
- SRT- ratio of the mass of biological solids in the system to that escaping from the reactor
- Max SRT, desirable for the stability of an anaerobic system and for minimizing the sludge production

HIGH RATE ANAEROBIC TREATMENT

- However, tend to increase the reactor volume , increasing the capital costs because minimal HRT minimizes the reactor volume.
- Other requirements
 - Intimate contact between the incoming wastewater and the detained biological solids in the reactor
 - Maintenance of sufficiently warm temperatures



■ Advantages

- Anaerobic processes give important advantage of generating valuable stream , biogas which can be used to generate electricity
- They also have the important advantage of not requiring energy for aeration

■ Disadvantages

- They have the disadvantage of a lower degradation rate of the organic substrate, therefore requiring larger vessels
- It is more difficult to achieve the same effluent quality in terms of COD removal with anaerobic digestion than with aerobic processes

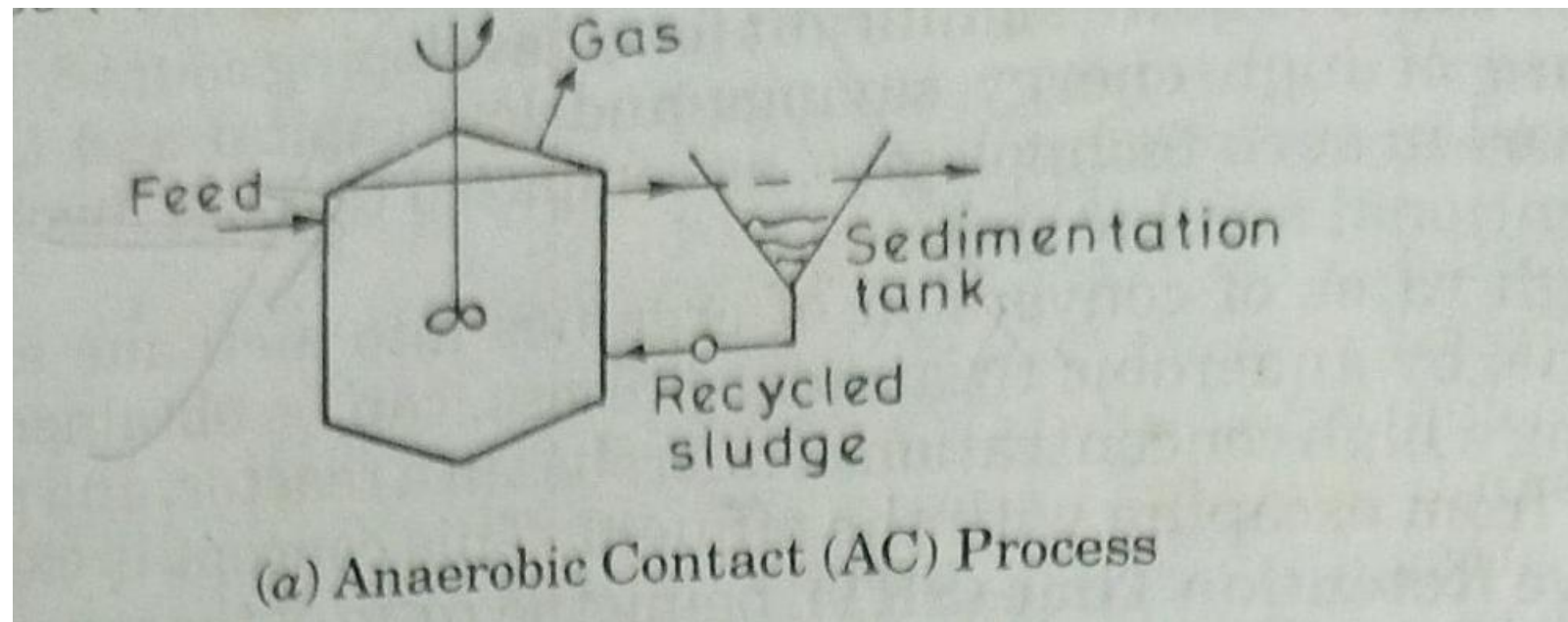
TYPES OF HIGH RATE ANAEROBIC SYSTEMS

- **Anaerobic contact process (AC) process**
- **Anaerobic filters (AF)**
- **Anaerobic fixed films (AFF) reactors**
- **Fluidized bed (FB) reactors**
- **Upflow anaerobic sludge blanket (UASB) reactor**

ANAEROBIC CONTACT PROCESS (AC) PROCESS

- System involves a closed stirred tank reactor in which the biomass leaving with the reactor effluent
- These are settled in the sedimentation tank and is recycled to the stirred tank, to increase SRT
- Continuous or complete mixing in the reactor is not recommended, since this may adversely affect the settling characteristics of the sludge
- Inadequate mixing may cause the formation of dead zones inside the reactor
- Generally used for treatment of industrial wastewater

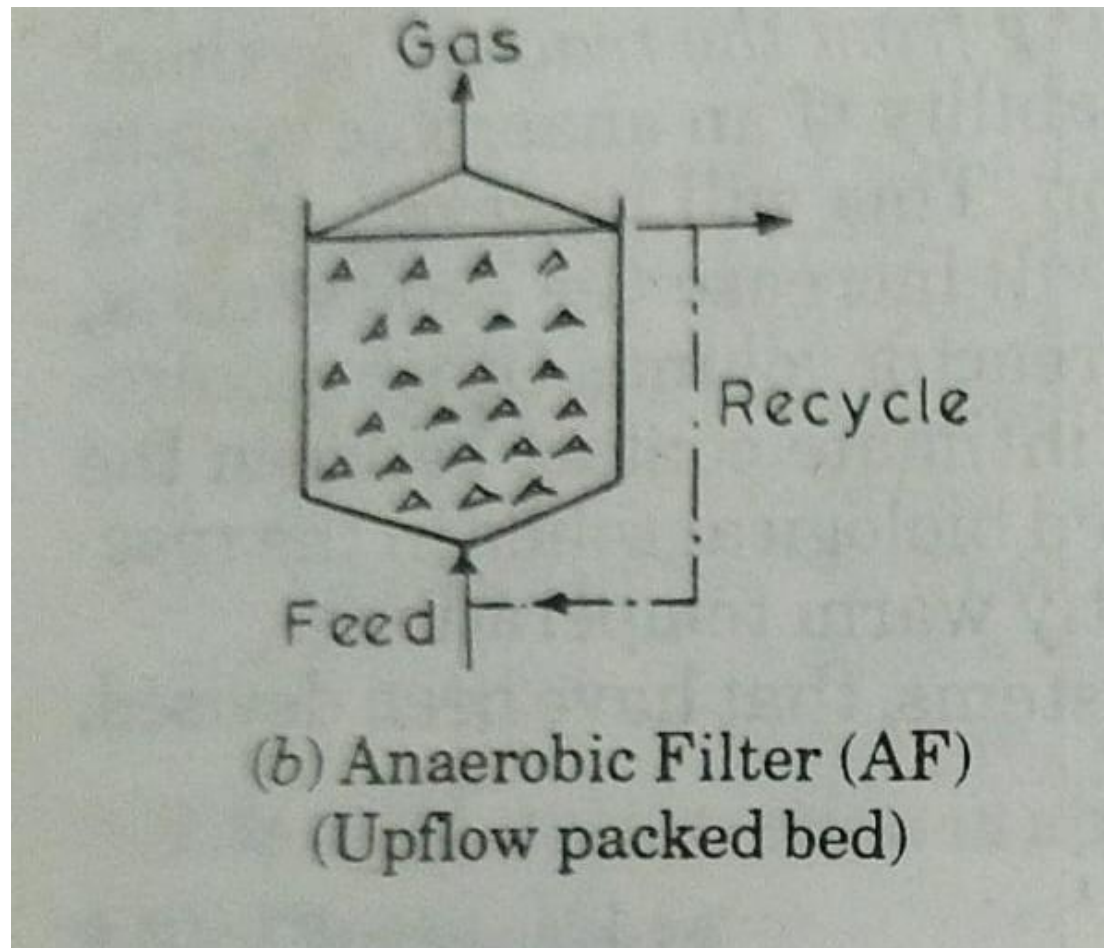
ANAEROBIC CONTACT PROCESS



ANAEROBIC FILTER

- A stationary filter media, consisting of crushed rock of 15-25mm size or plastic beads etc. packed in a closed tank
- Wastewater entering from the bottom move up the packing media
- The sludge is entrapped in the void space between the packing material
- The effluent is recycled to increase SRT of the reactor
- Constructed for the treatment of low strength municipal wastewater

ANAEROBIC FILTER

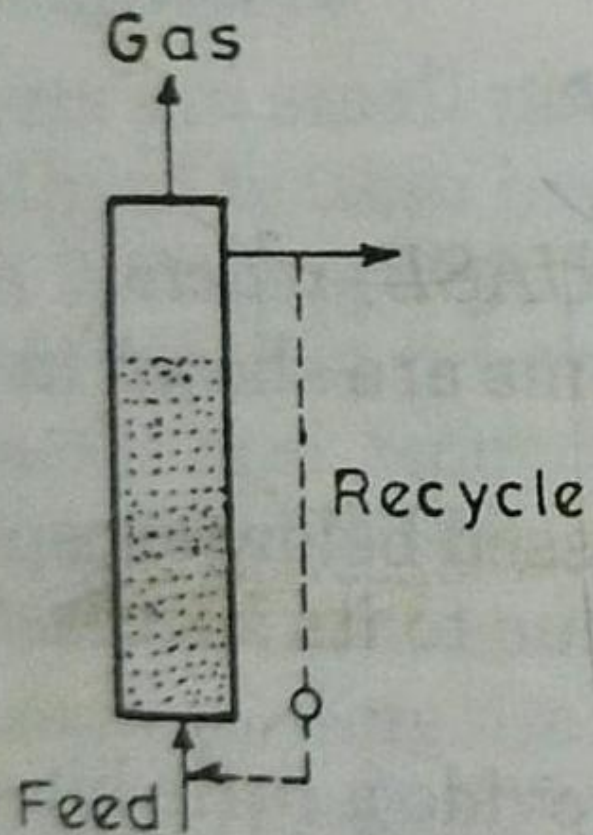


ANAEROBIC FIXED FILMS REACTOR

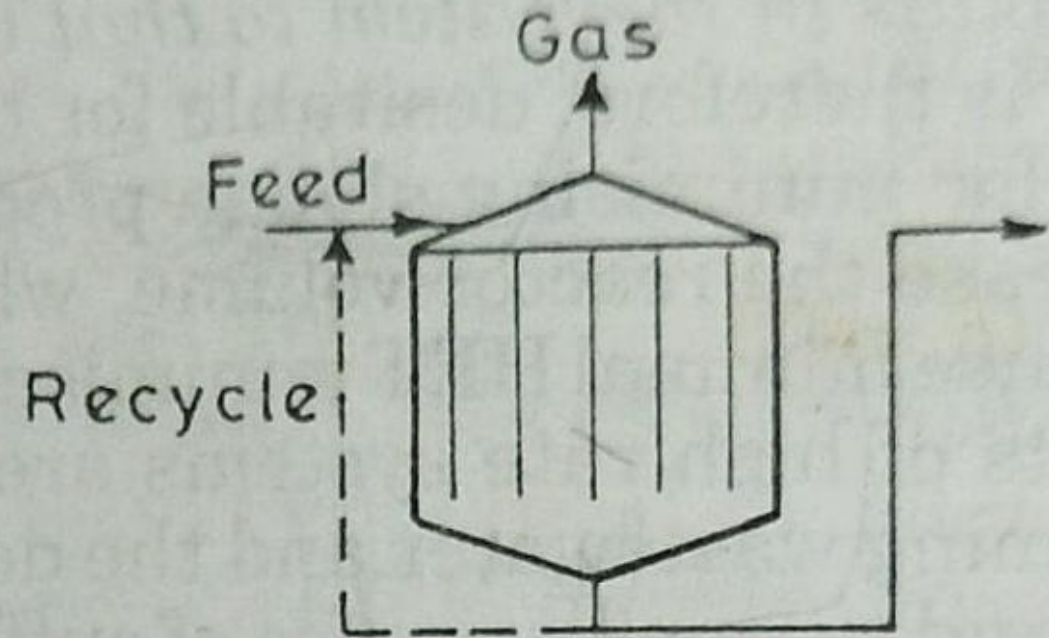
- This is also characterized by the presence of stationary packing material in the reactor
- The aim is to avoid entrapment of suspended solids in the voids of the packing material
- To prevent accumulation of solids in the reactor, it is worked with downflow mode
- Can be operated either in submerged or unsubmerged conditions
- Packing is usually of modular construction , consists of plastic sheets , providing a high void ratio
- Used to treat high strength waste , containing high amount of suspended solids

FLUIDISED AND EXPANDED BED REACTORS

- Characterised by the presence of mobile packing material, such as clay, sand, coal etc
- Organics and microbes in the reactor get attached to these particles , with which the reactor tank is filled only to a part height
- Wastewater is entered from the bottom of the tank with an upflow velocity to move the media particles which act as mobile biomass carriers , causing digestion of the organics present in the wastewater
- Fluidised bed – overflow velocity is high, causing entire bed material to move with the up moving sewage, thereby disturbing the sludge
- Expanded bed – do not aim at complete fluidization of the bed , permitting the use of lower upflow pressure on the wastewater , thereby consuming lesser electrical power. Can be used for treating industrial as well as municipal wastewater



(d) Fluidized/Expanded
Bed reactor



(c) Anaerobic fixed films (AFF)
reactor (Downflow packed bed)

UPFLOW ANAEROBIC SLUDGE BLANKET REACTOR--- UASB UNITS

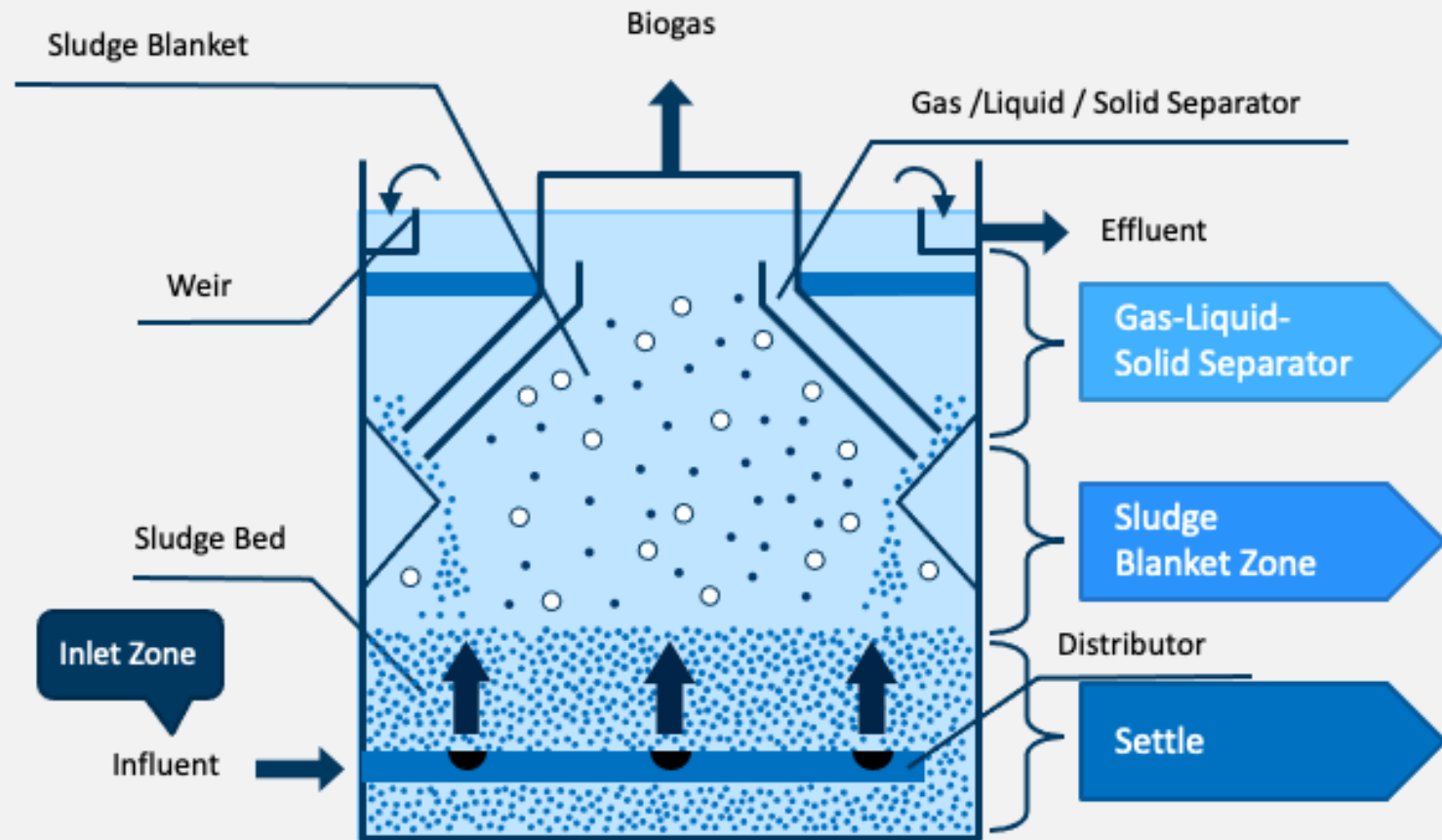
- **UASB type units** are one in which no special media have to be used since the sludge granules themselves act as the 'media' and stay in suspension
- Arrangement of a USAB type treatment plant
 - ❑ Initial pumping
 - ❑ Screening and degritting
 - ❑ Main UASB reactor
 - ❑ Gas collection and conversion or conveyance
 - ❑ Sludge drying bed
 - ❑ Post treatment facility

UASB UNITS

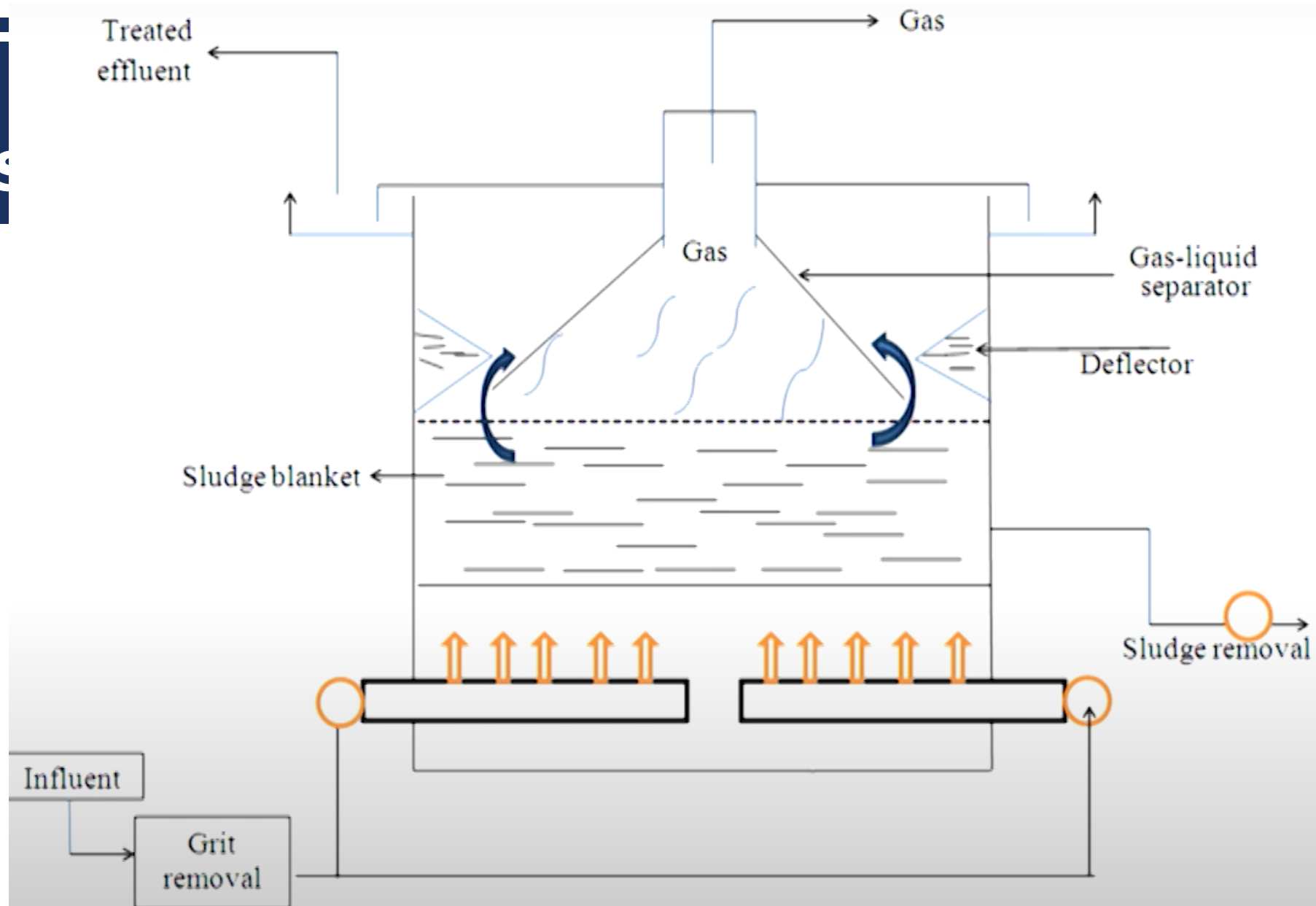
UP FLOW ANAEROBIC SLUDGE BLANKET

Up Flow Anaerobic Sludge Blanket (UASB) Reactor

Source-Industrial technology research institute



UAS



Schematic diagram of up-flow anaerobic sludge blanket (UASB)

UASB UNITS

- In the UASB process, the whole waste is passed through the anaerobic reactor in an upflow mode, with a hydraulic retention time (HRT) of only about 8-10 hours at average flow.
- No prior sedimentation is required
- Anaerobic unit does not need to be filled with stones or any other media; the upflowing sewage itself forms millions of small "granules" or particles of sludge which are held in suspension and provide a large surface area on which organic matter can attach and undergo biodegradation
- A high solid retention time (SRT) of 30-50 or more days occurs within the unit

UASB UNITS

- No mixers or aerators are required
- Gas produced can be collected and used if desired
- Anaerobic systems function satisfactorily when temperatures inside the reactor are above 18-20°C.
- Excess sludge is removed from time to time through a separate pipe and sent to a simple sand bed for drying.

ADVANTAGES OF UASB SYSTEM

- Advantages of USAB over conventional aerobic system are
 - Space requirement of the system is quite comparable to that an ASP, ie about 0.5 acres per MLD , as compared to 2.5 acres per MLD required for oxidation ponds , and acres for aerated lagoons
 - Capital cost investment of such plant is about Rs.20 Lakhs/MLD as compared to about Rs. 35 lakhs/MLD for an ASP, Rs 7.5 lakh/MLD for oxidation ponds , and Rs.15 lakhs/MLD for aerated lagoon
 - The system requires lesser and simpler electromagnetic parts as compared to the ones required in an ASP, leading to lower O and M costs
 - Electricity consumption in this system, like all anaerobic systems, is quite low, and the system is quite capable of withstanding long power failures
 - The sludge production in the system , and the produced sludge is having quick dewatering characteristics
 - The system enables quicker sludge digestion , as compared to the conventional digestors
 - Biogas is produced in the system as a by-product , which can be used to produce electricity to run the system

LIMITATIONS OR DRAWBACKS OF USAB SYSTEM

- Drawbacks of USAB over the conventional aerobic systems are
 - Takes a longer start-up time to establish the steady-state condition
 - Requires further treatment with an aerobic treatment process to meet the discharge requirements in many cases
 - Doesn't remove nitrogen and phosphorus biologically
 - Process is affected when toxic substances are present in the feed
 - Has potential for production of odour and corrosive gases

APPLICATIONS

- The UASB is employed for the treatment of industrial wastes and the treatment of primary and secondary sludge
- UASB has been found very effective for the treatment of municipal wastewater
- UASB process is followed by the post-treatment by other biological processes such as ponds and lagoons
- UASB process is considered one of the important options for sewage treatment in warm countries